

THE EIGHTEENTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

February 8, 1958

Morning Session

1. If $a_0, a_1, \dots, a_n$ are real numbers satisfying

$$\frac{a_0}{1} + \frac{a_1}{2} + \dots + \frac{a_n}{n+1} = 0,$$

show that the equation $a_0 + a_1x + a_2x^2 + \dots + a_nx^n = 0$ has at least one real root. (page 454)

2. Two uniform solid spheres of equal radii are so placed that one is directly above the other. The bottom sphere is fixed, and the top sphere, initially at rest, rolls off. At what point will contact between the two spheres be "lost"? Assume the coefficient of friction is such that no slipping occurs. (page 454)

3. Real numbers are chosen at random from the interval $(0 \leq x \leq 1)$. If after choosing the n th number the sum of the numbers so chosen first exceeds 1, show that the expected or average value for n is e . (page 457)

4. If $a_1, a_2, \dots, a_n$ are complex numbers such that

$$|a_1| = |a_2| = \dots = |a_n| = r \neq 0,$$

and if ${}_nT_s$ denotes the sum of all products of these n numbers taken s at a time, prove that

$$\left| \frac{{}_nT_s}{{}_nT_{n-s}} \right| = r^{2s-n}$$

whenever the denominator of the left-hand side is different from zero. (page 458)

5. Show that the integral equation

$$f(x, y) = 1 + \int_0^x \int_0^y f(u, v) du dv$$

has at most one solution continuous for $0 \leq x \leq 1, 0 \leq y \leq 1$. (page 459)

6. What is the smallest amount that may be invested at interest rate i , compounded annually, in order that one may withdraw 1 dollar at the end of the first year, 4 dollars at the end of the second year, $\dots$, n^2 dollars at the end of the n th year, in perpetuity? (page 460)

7. Show that ten equal-sized squares cannot be placed on a plane in such a way that no two have an interior point in common and the first touches each of the others. (page 461)

Afternoon Session

1. (i) Given line segments A, B, C, D , with A the longest, construct a quadrilateral with these sides and with A and B parallel, when possible.

(ii) Given any acute-angled triangle ABC and one altitude AH , select any point D on AH , then draw BD and extend until it intersects AC in E , and draw CD and extend until it intersects AB in F . Prove angle $AHE =$ angle AHF . (page 463)

2. Prove that the product of four consecutive positive integers cannot be a perfect square or cube. (page 466)

3. In a round-robin tournament with n players (each pair of players plays one game) in which there are no draws, the numbers of wins scored by the players are $s_1, s_2, \dots, s_n$. Prove that a necessary and sufficient condition for the existence of 3 players, A, B, C , such that A beat B , B beat C , and C beat A is

$$s_1^2 + s_2^2 + \dots + s_n^2 < (n-1)(n)(2n-1)/6. \quad (\text{page 467})$$

4. What is the average straight line distance between two points on a sphere of radius 1? (page 468)

5. Given an infinite number of points in a plane, prove that if all the distances determined between them are integers then the points are all in a straight line. (page 469)

6. A projectile moves in a resisting medium. The resisting force is a function of the velocity and is directed along the velocity vector. The equation $x = f(t)$ gives the horizontal distance in terms of the time t . Show that the vertical distance y is given by

$$y = -gf(t) \int \frac{dt}{f'(t)} + g \int \frac{f(t)}{f'(t)} dt + Af(t) + B$$

where A and B are constants and g is the acceleration due to gravity. (page 470)

7. Prove that if $f(x)$ is continuous for $a \leq x \leq b$ and $\int_a^b x^n f(x) dx = 0$ for $n = 0, 1, 2, \dots$ then $f(x)$ is identically zero on $a \leq x \leq b$. (page 472)

**THE NINETEENTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

November 22, 1958

Morning Session

1. Let $f(m, 1) = f(1, n) = 1$ for $m \geq 1, n \geq 1$, and let $f(m, n) = f(m-1, n) + f(m, n-1) + f(m-1, n-1)$ for $m > 1$ and $n > 1$. Also let

$$S(n) = \sum_{a+b=n} f(a, b), \quad a \geq 1 \text{ and } b \geq 1.$$

Prove that

$$S(n+2) = S(n) + 2S(n+1) \quad \text{for } n \geq 2.$$

(page 476)

2. Let

$$R_1 = 1, \quad R_{n+1} = 1 + n/R_n, \quad n \geq 1.$$

Show that for $n \geq 1$,

$$\sqrt{n} \leq R_n \leq \sqrt{n} + 1. \quad \text{(page 477)}$$

3. Under the assumption that the following set of relations has a unique solution for $u(t)$, determine it.

$$\frac{du(t)}{dt} = u(t) + \int_0^1 u(s) ds,$$

$$u(0) = 1. \quad \text{(page 478)}$$

4. In assigning dormitory rooms, a college gives preference to pairs of students in this order:

$$AA, AB, AC, BB, BC, AD, CC, BD, CD, DD,$$

in which AA means two seniors, AB means a senior and a junior, etc. Determine numerical values to assign to A, B, C, D so that the set of numbers A

$+A, A+B, A+C, B+B$, etc., corresponding to the order above will be in descending magnitude. Find the general solution and the solution in least positive integers. (page 480)

5. Show that the number of non-zero terms in the expansion of the n th order determinant having zeros in the main diagonal and ones elsewhere is

$$n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!} \right]. \quad (\text{page 481})$$

6. Let $a(x)$ and $b(x)$ be continuous functions on $0 \leq x \leq 1$ and let $0 \leq a(x) \leq a < 1$ on that range. Under what other conditions (if any) is the solution of the equation for u ,

$$u = \text{maximum}_{0 \leq x \leq 1} [b(x) + a(x) \cdot u],$$

given by

$$u = \text{maximum}_{0 \leq x \leq 1} \left[\frac{b(x)}{1 - a(x)} \right]? \quad (\text{page 485})$$

7. Let a and b be relatively prime positive integers, b even. For each positive integer q let $p = p(q)$ be chosen so that

$$\left| \frac{p}{q} - \frac{a}{b} \right|$$

is a minimum. Prove that

$$\lim_{n \rightarrow \infty} \sum_{q=1}^n \frac{q \left| \frac{p}{q} - \frac{a}{b} \right|}{n} = \frac{1}{4}. \quad (\text{page 486})$$

Afternoon Session

1. Given

$$b_n = \sum_{k=0}^n \binom{n}{k}^{-1}, \quad n \geq 1,$$

prove that

$$b_n = \frac{n+1}{2n} b_{n-1} + 1, \quad n \geq 2,$$

and hence, as a corollary,

$$\lim_{n \rightarrow \infty} b_n = 2. \quad (\text{page 487})$$

2. Given a set of $n + 1$ positive integers, none of which exceeds $2n$, show that at least one member of the set must divide another member of the set. (page 489)

3. If a square of unit side be partitioned into two sets, then the diameter (least upper bound of the distances between pairs of points) of one of the sets is not less than $\sqrt{5}/2$. Show also that no larger number will do. (page 489)

4. Let C be a real number, and let f be a function such that

$$\lim_{x \rightarrow \infty} f(x) = C, \quad \lim_{x \rightarrow \infty} f'''(x) = 0.$$

Prove that

$$\lim_{x \rightarrow \infty} f'(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} f''(x) = 0,$$

where superscripts denote derivatives. (page 490)

5. The lengths of successive segments of a broken line are represented by the successive terms of the harmonic progression $1, 1/2, 1/3, \dots, 1/n, \dots$. Each segment makes with the preceding segment a given angle θ . What is the distance and what is the direction of the limiting point (if there is one) from the initial point of the first segment? (page 491)

6. Let a complete oriented graph on n points be given, i.e., a set of n points $1, 2, 3, \dots, n$, and between any two points i and j a direction, $i \rightarrow j$. Show that there exists a permutation of the points, $[a_1, a_2, a_3, \dots, a_n]$, such that $a_1 \rightarrow a_2 \rightarrow a_3 \rightarrow \dots \rightarrow a_n$. (page 494)

7. Let $a_1, a_2, \dots, a_n$ be a permutation of the integers $1, 2, \dots, n$. Call a_i a "big" integer if $a_i > a_j$ for all $j > i$. Find the mean number of "big" integers over all permutations on the first n positive integers. (page 495)